

GOC/Host PC Upgrade to GOC with T5810



OPERATING DOCUMENTATION

5670090
Revision 4.0

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Important Information

LANGUAGE

ПРЕДУПРЕЖДЕНИЕ (BG)

Това упътване за работа е налично само на английски език.

- Ако доставчикът на услугата на клиента изиска друг език, задължение на клиента е да осигури превод.
- Не използвайте оборудването, преди да сте се консултирали и разбрали упътването за работа.
- Неспазването на това предупреждение може да доведе до нараняване на доставчика на услугата, оператора или пациента в резултат на токов удар, механична или друга опасност.

警告 (ZH-CN)

本维修手册仅提供英文版本。

- 如果客户的维修服务人员需要非英文版本，则客户需自行提供翻译服务。
- 未详细阅读和完全理解本维修手册之前，不得进行维修。
- 忽略本警告可能对维修服务人员、操作人员或患者造成电击、机械伤害或其他形式的伤害。

警告 (ZH-HK)

本服務手冊僅提供英文版本。

- 倘若客戶的服務供應商需要英文以外之服務手冊，客戶有責任提供翻譯服務。
- 除非已參閱本服務手冊及明白其內容，否則切勿嘗試維修設備。
- 不遵從本警告或會令服務供應商、網絡供應商或病人受到觸電、機械性或其他危險。

警告 (ZH-TW)

本維修手冊僅有英文版。

- 若客戶的維修廠商需要英文版以外的語言，應由客戶自行提供翻譯服務。
- 請勿試圖維修本設備，除非 您已查閱並瞭解本維修手冊。
- 若未留意本警告，可能導致維修廠商、操作員或病患因觸電、機械或其他危險而受傷。

UPOZORENJE (HR)

Ovaj servisni priručnik dostupan je na engleskom jeziku.

- Ako davatelj usluge klijenta treba neki drugi jezik, klijent je dužan osigurati prijevod.
- Ne pokušavajte servisirati opremu ako niste u potpunosti pročitali i razumjeli ovaj servisni priručnik.
- Zanimarite li ovo upozorenje, može doći do ozljede davatelja usluge, operatera ili pacijenta uslijed strujnog udara, mehaničkih ili drugih rizika.

**VÝSTRAHA
(CS)**

Tento provozní návod existuje pouze v anglickém jazyce.

- V případě, že externí služba zákazníkům potřebuje návod v jiném jazyce, je zajištění překladu do odpovídajícího jazyka úkolem zákazníka.
- Nesnažte se o údržbu tohoto zařízení, aniž byste si přečetli tento provozní návod a pochopili jeho obsah.
- V případě nedodržování této výstrahy může dojít k poranění pracovníka prodejního servisu, obslužného personálu nebo pacientů vlivem elektrického proudu, respektive vlivem mechanických či jiných rizik.

**ADVARSEL
(DA)**

Denne servicemanual findes kun på engelsk.

- Hvis en kundes tekniker har brug for et andet sprog end engelsk, er det kundens ansvar at sørge for oversættelse.
- Forsøg ikke at servicere udstyret uden at læse og forstå denne servicemanual.
- Manglende overholdelse af denne advarsel kan medføre skade på grund af elektrisk stød, mekanisk eller anden fare for teknikeren, operatøren eller patienten.

**WAARSCHUWING
(NL)**

Deze onderhoudshandleiding is enkel in het Engels verkrijgbaar.

- Als het onderhoudspersoneel een andere taal vereist, dan is de klant verantwoordelijk voor de vertaling ervan.
- Probeer de apparatuur niet te onderhouden alvorens deze onderhoudshandleiding werd geraadpleegd en begrepen is.
- Indien deze waarschuwing niet wordt opgevolgd, zou het onderhoudspersoneel, de operator of een patiënt gewond kunnen raken als gevolg van een elektrische schok, mechanische of andere gevaren.

**WARNING
(EN)**

This service manual is available in English only.

- If a customer's service provider requires a language other than English, it is the customer's responsibility to provide translation services.
- Do not attempt to service the equipment unless this service manual has been consulted and is understood.
- Failure to heed this warning may result in injury to the service provider, operator or patient from electric shock, mechanical or other hazards.

**HOIATUS
(ET)**

See teenindusjuhend on saadaval ainult inglise keeles.

- Kui klienditeeninduse osutaja nõuab juhendit inglise keelest erinevas keeles, vastutab klient tõlketeenuse osutamise eest.
- Ärge üritage seadmeid teenindada enne eelnevalt käesoleva teenindusjuhendiga tutvumist ja sellest aru saamist.
- Käesoleva hoiatuse eiramine võib põhjustada teenuseosutaja, operaatori või patsiendi vigastamist elektrilöögi, mehaanilise või muu ohu tagajärjel.

**VAROITUS
(FI)**

Tämä huolto-ohje on saatavilla vain englanniksi.

- Jos asiakkaan huoltohenkilöstö vaatii muuta kuin englanninkielistä materiaalia, tarvittavan käännöksen hankkiminen on asiakkaan vastuulla.
- Älä yritä korjata laitteistoa ennen kuin olet varmasti lukenut ja ymmärtänyt tämän huolto-ohjeen.
- Mikäli tätä varoitusta ei noudateta, seurauksena voi olla huoltohenkilöstön, laitteen käyttäjän tai potilaan vahingoittuminen sähköiskun, mekaanisen vian tai muun vaaratilanteen vuoksi.

**ATTENTION
(FR)**

Ce manuel d'installation et de maintenance est disponible uniquement en anglais.

- Si le technicien d'un client a besoin de ce manuel dans une langue autre que l'anglais, il incombe au client de le faire traduire.
- Ne pas tenter d'intervenir sur les équipements tant que ce manuel d'installation et de maintenance n'a pas été consulté et compris.
- Le non-respect de cet avertissement peut entraîner chez le technicien, l'opérateur ou le patient des blessures dues à des dangers électriques, mécaniques ou autres.

**WARNUNG
(DE)**

Diese Serviceanleitung existiert nur in englischer Sprache.

- Falls ein fremder Kundendienst eine andere Sprache benötigt, ist es Aufgabe des Kunden für eine entsprechende Übersetzung zu sorgen.
- Versuchen Sie nicht diese Anlage zu warten, ohne diese Serviceanleitung gelesen und verstanden zu haben.
- Wird diese Warnung nicht beachtet, so kann es zu Verletzungen des Kundendienst-technikers, des Bedieners oder des Patienten durch Stromschläge, mechanische oder sonstige Gefahren kommen.

**ΠΡΟΕΙΔΟΠΟΙΗΣΗ
(EL)**

Το παρόν εγχειρίδιο σέρβις διατίθεται μόνο στα αγγλικά.

- Εάν ο τεχνικός σέρβις ενός πελάτη απαιτεί το παρόν εγχειρίδιο σε γλώσσα εκτός των αγγλικών, αποτελεί ευθύνη του πελάτη να παρέχει τις υπηρεσίες μετάφρασης.
- Μην επιχειρήσετε την εκτέλεση εργασιών σέρβις στον εξοπλισμό αν δεν έχετε συμβουλευτεί και κατανοήσει το παρόν εγχειρίδιο σέρβις.
- Αν δεν προσέξετε την προειδοποίηση αυτή, ενδέχεται να προκληθεί τραυματισμός στον τεχνικό σέρβις, στο χειριστή ή στον ασθενή από ηλεκτροπληξία, μηχανικούς ή άλλους κινδύνους.

**FIGYELMEZTETÉS
(HU)**

Ezen karbantartási kézikönyv kizárólag angol nyelven érhető el.

- Ha a vevő szolgáltatója angoltól eltérő nyelvre tart igényt, akkor a vevő felelőssége a fordítás elkészítése.
- Ne próbálja elkezdni használni a berendezést, amíg a karbantartási kézikönyvben leírtakat nem értelmezték.
- Ezen figyelmeztetés figyelmen kívül hagyása a szolgáltató, működtető vagy a beteg áramütés, mechanikai vagy egyéb veszélyhelyzet miatti sérülését eredményezheti.

**AÐVÖRUN
(IS)**

Þessi þjónustuhandbók er aðeins fáanleg á ensku.

- Ef að þjónustuveitandi viðskiptamanns þarfnast annas tungumáls en ensku, er það skylda viðskiptamanns að skaffa tungumálaþjónustu.
- Reynið ekki að afgreiða tækið nema að þessi þjónustuhandbók hefur verið skoðuð og skilin.
- Brot á sinna þessari aðvörun getur leitt til meiðsla á þjónustuveitanda, stjórnanda eða sjúklings frá raflosti, vélrænu eða öðrum áhættum.

**AVVERTENZA
(IT)**

Il presente manuale di manutenzione è disponibile soltanto in lingua inglese.

- Se un addetto alla manutenzione richiede il manuale in una lingua diversa, il cliente è tenuto a provvedere direttamente alla traduzione.
- Procedere alla manutenzione dell'apparecchiatura solo dopo aver consultato il presente manuale ed averne compreso il contenuto.
- Il mancato rispetto della presente avvertenza potrebbe causare lesioni all'addetto alla manutenzione, all'operatore o ai pazienti provocate da scosse elettriche, urti meccanici o altri rischi.

**警告
(JA)**

このサービスマニュアルには英語版しかありません。

- サービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。
- このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないでください。
- この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

**경고
(KO)**

본 서비스 매뉴얼은 영어로만 이용하실 수 있습니다.

- 고객의 서비스 제공자가 영어 이외의 언어를 요구할 경우, 번역 서비스를 제공하는 것은 고객의 책임입니다.
- 본 서비스 매뉴얼을 참조하여 숙지하지 않은 이상 해당 장비를 수리하려고 시도하지 마십시오.
- 본 경고 사항에 유의하지 않으면 전기 쇼크, 기계적 위험, 또는 기타 위험으로 인해 서비스 제공자, 사용자 또는 환자에게 부상을 입힐 수 있습니다.

**BRĪDINĀJUMS
(LV)**

Šī apkopes rokasgrāmata ir pieejama tikai angļu valodā.

- Ja klienta apkopes sniedzējam nepieciešama informācija citā valodā, klienta pienākums ir nodrošināt tulkojumu.
- Neveiciet aprikojuma apkopi bez apkopes rokasgrāmatas izlasīšanas un saprašanas.
- Šī brīdinājuma neievērošanas rezultātā var rasties elektriskās strāvas trieciena, mehānisku vai citu faktoru izraisītu traumu risks apkopes sniedzējam, operatoram vai pacientam.

**[SPĖJIMAS
(LT)]**

Šis eksploatavimo vadovas yra tik anglų kalba.

- Jei kliento paslaugų tiekėjas reikalauja vadovo kita kalba – ne anglų, suteikti vertimo paslaugas privalo klientas.
- Nemėginkite atlikti įrangos techninės priežiūros, jei neperskaitėte ar nesupratote šio eksploatavimo vadovo.
- Jei nepaisysite šio įspėjimo, galimi paslaugų tiekėjo, operatoriaus ar paciento sužalojimai dėl elektros šoko, mechaninių ar kitų pavojų.

**ADVARSEL
(NO)**

Denne servicehåndboken finnes bare på engelsk.

- Hvis kundens serviceleverandør har bruk for et annet språk, er det kundens ansvar å sørge for oversettelse.
- Ikke forsøk å reparere utstyret uten at denne servicehåndboken er lest og forstått.
- Manglende hensyn til denne advarselen kan føre til at serviceleverandøren, operatøren eller pasienten skades på grunn av elektrisk støt, mekaniske eller andre farer.

**OSTRZEŻENIE
(PL)**

Niniejszy podręcznik serwisowy dostępny jest jedynie w języku angielskim.

- Jeśli serwisant klienta wymaga języka innego niż angielski, zapewnienie usługi tłumaczenia jest obowiązkiem klienta.
- Nie próbować serwisować urządzenia bez zapoznania się z niniejszym podręcznikiem serwisowym i zrozumienia go.
- Niezastosowanie się do tego ostrzeżenia może doprowadzić do obrażeń serwisanta, operatora lub pacjenta w wyniku porażenia prądem elektrycznym, zagrożenia mechanicznego bądź innego.

**ATENÇÃO
(PT-BR)**

Este manual de assistência técnica encontra-se disponível unicamente em inglês.

- Se outro serviço de assistência técnica solicitar a tradução deste manual, caberá ao cliente fornecer os serviços de tradução.
- Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica.
- A não observância deste aviso pode ocasionar ferimentos no técnico, operador ou paciente decorrentes de choques elétricos, mecânicos ou outros.

**ATENÇÃO
(PT-PT)**

Este manual de assistência técnica só se encontra disponível em inglês.

- Se qualquer outro serviço de assistência técnica solicitar este manual noutro idioma, é da responsabilidade do cliente fornecer os serviços de tradução.
- Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica.
- O não cumprimento deste aviso pode colocar em perigo a segurança do técnico, do operador ou do paciente devido a choques eléctricos, mecânicos ou outros.

**ATENȚIE
(RO)**

Acest manual de service este disponibil doar în limba engleză.

- Dacă un furnizor de servicii pentru clienți necesită o altă limbă decât cea engleză, este de datoria clientului să furnizeze o traducere.
- Nu încercați să reparați echipamentul decât ulterior consultării și înțelegerii acestui manual de service.
- Ignorarea acestui avertisment ar putea duce la rănirea depanatorului, operatorului sau pacientului în urma pericolelor de electrocutare, mecanice sau de altă natură.

**ОСТОРОЖНО!
(RU)**

Данное руководство по техническому обслуживанию представлено только на английском языке.

- Если сервисному персоналу клиента необходимо руководство не на английском, а на каком-то другом языке, клиенту следует самостоятельно обеспечить перевод.
- Перед техническим обслуживанием оборудования обязательно обратитесь к данному руководству и поймите изложенные в нем сведения.
- Несоблюдение требований данного предупреждения может привести к тому, что специалист по техобслуживанию, оператор или пациент получит удар электрическим током, механическую травму или другое повреждение.

**UPOZORENJE
(SR)**

Ovo servisno uputstvo je dostupno samo na engleskom jeziku.

- Ako klijentov serviser zahteva neki drugi jezik, klijent je dužan da obezbedi prevodilačke usluge.
- Ne pokušavajte da opravite uređaj ako niste pročitali i razumeli ovo servisno uputstvo.
- Zanemarivanje ovog upozorenja može dovesti do povređivanja serviser, rukovaoca ili pacijenta usled strujnog udara ili mehaničkih i drugih opasnosti.

**UPOZORNENIE
(SK)**

Tento návod na obsluhu je k dispozícii len v angličtine.

- Ak zákazníkovi poskytovateľ služieb vyžaduje iný jazyk ako angličtinu, poskytnutie prekladateľských služieb je zodpovednosťou zákazníka.
- Nepokúšajte sa o obsluhu zariadenia, kým si neprečítate návod na obsluhu a neporozumiete mu.
- Zanedbanie tohto upozornenia môže spôsobiť zranenie poskytovateľa služieb, obsluhujúcej osoby alebo pacienta elektrickým prúdom, mechanické alebo iné ohrozenie.

**ATENCION
(ES)**

Este manual de servicio sólo existe en inglés.

- Si el encargado de mantenimiento de un cliente necesita un idioma que no sea el inglés, el cliente deberá encargarse de la traducción del manual.
- No se deberá dar servicio técnico al equipo, sin haber consultado y comprendido este manual de servicio.
- La no observancia del presente aviso puede dar lugar a que el proveedor de servicios, el operador o el paciente sufran lesiones provocadas por causas eléctricas, mecánicas o de otra naturaleza.

**VARNING
(SV)**

Den här servicehandboken finns bara tillgänglig på engelska.

- Om en kunds servicetekniker har behov av ett annat språk än engelska, ansvarar kunden för att tillhandahålla översättningstjänster.
- Försök inte utföra service på utrustningen om du inte har läst och förstår den här servicehandboken.
- Om du inte tar hänsyn till den här varningen kan det resultera i skador på serviceteknikern, operatören eller patienten till följd av elektriska stötar, mekaniska faror eller andra faror.

**OPOZORILO
(SL)**

Ta servisni priročnik je na voljo samo v angleškem jeziku.

- Če ponudnik storitve stranke potrebuje priročnik v drugem jeziku, mora stranka zagotoviti prevod.
- Ne poskušajte servisirati opreme, če tega priročnika niste v celoti prebrali in razumeli.
- Če tega opozorila ne upoštevate, se lahko zaradi električnega udara, mehanskih ali drugih nevarnosti poškoduje ponudnik storitev, operater ali bolnik.

**DİKKAT
(TR)**

Bu servis kılavuzunun sadece ingilizcesi mevcuttur.

- Eğer müşteri teknisyeni bu kılavuzu ingilizce dışında bir başka lisandan talep ederse, bunu tercüme ettirmek müşteriye düşer.
- Servis kılavuzunu okuyup anlamadan ekipmanlara müdahale etmeyiniz.
- Bu uyarıya uyulmaması, elektrik, mekanik veya diğer tehlikelerden dolayı teknisyen, operatör veya hastanın yaralanmasına yol açabilir.

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Revision History

Revision	Date	Description
1.0	June 16, 2014	Initial release.
2.0	October 22, 2014	Section 1.9.5: Completed reference link. Section 1.4.3: Added detail that ethernet Run 3011 should connect to the right side port at rear of z420 PC.
3.0	June 4, 2016	Chapter 1, Section 1.1: Updated T5810 host compatibility with DV25.1 and later only. Chapter 1, Section 1.9.2: Updated contact information for return of recycled components. Chapter 2, Section 1.3: Updated upgrade procedures for a GOC with a Dell T5810. Chapter 3, Section 1.2: Updated information from z420 to T5810 compatible with DV25.1 and later only.
4.0	February 23, 2018	Chapter 1: Added instructions that 8400 requires complete GOC replacement. Revised harvest information. Added instructions for MyPLCReader App. Chapter 2: Added note that 8400 requires complete GOC replacement. Updated illustrations and instructions to indicate that later GOCs do not include the 8-port Ethernet hub. Chapter 3: Added USB mouse fix procedure.

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Chapter 1 Overview

1 Getting Started

1.1 Upgrade Introduction

This publication provides instructions to upgrade an existing GOC with an 8400, z400, or z420 Host PC to a GOC with a Dell T5810.

(For GOC with 8400 Host PC) The entire GOC must be completely removed and replaced with a new GOC containing the Dell T5810. Refer to the *GOC Upgrade Procedures* section of this manual.

(For GOC with z400 or z420 Host PC) Either the entire GOC can be replaced or just the Host PC can be upgraded. To upgrade the entire GOC, refer to the *GOC Upgrade Procedures* section of this manual. To upgrade only the Host PC, refer to the *Dell T5810 Computer Upgrade Replacement* procedures in the Service Manual. Follow all other instructions as provided in the procedures in this manual to ensure satisfactory completion of Harvest and Product locator activities.

The T5810 Host PC is only compatible with DV25.1 and later software.



WARNING

FERROUS MATERIAL HAZARD!

DELIVERY EQUIPMENT AND OTHER TOOLS AND PARTS REQUIRED FOR THIS INSTALLATION CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES.

KEEP ALL FERROUS TOOLS AT LEAST 10 FEET AWAY FROM THE MAGNET.

1.2 Site Ready Check

Pre-installation work must be completed before equipment is delivered to avoid delays and confusion.

1.3 Pre-Delivery FDO Checks

Before the shipment is made, Be sure that FDO review has been completed by consultation with the Upgrade Specialist. Also verify the Surface coil compatibility issues have also been resolved.

1.4 Upgrade Schedule Considerations

FMLs, periodic maintenance, installation of prerequisite upgrades and/or options will directly add to the man-hour and system down-time total. Be sure that all planned activity is considered when predicting the return of system to customer.

1.5 Product Delivery Instructions and Packing

The "Shipping Document" lists catalog numbers delivered. **Review and confirm that order is delivered complete.** Check impact on installation schedule if Catalogs and/or packing boxes are missing and/or noted as shipped short.

Labels that are attached to the outside of packing boxes summarize box contents. PDI (Product Delivery Instructions) specify box contents, part numbers, and shipping procedure. The PDI is numbered according to the catalog number. Lists of items included with each box are detailed by separate checklists, or a separate sheet that provides a summary of that box contents. Refer to PDI and packing lists for information specific to your shipment.

A set of service and operator manuals is delivered with the Upgrade Kit. Refer to checklists packed with "Technical Publication" boxes for a list of delivered documents.

1.6 Damage in Transportation

All packages should be closely examined at time of delivery. If damage is apparent, have notation "**damage in shipment**" written on **all** copies of the freight or express bill **before** delivery is accepted or "signed for" by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14** day period. File a report with:

- Call 1-800-548-3366. Select "Install Support Services for FOA and MIS"
- Contact your local service coordinator for more information on this process.

1.7 Product Locator

During installation, the Install FE shall confirm that the Model and Serial Numbers are part of the asset. If missing, use either the gib web interface (http://gib.gehealthcare.com/gib/gib_entry.jsp) or the MyPLCReader App. This information will then be loaded directly to Siebel by the IB Verification Team.

Use the following procedure to install the MyPLCReader App.

1. From the GE App Icon, search for and install the MyPLCReader App on your iPhone or iPad.

Illustration 1-1: MyPLCReader App



2. Update the Configuration Page.
 - a. The first time the App opens, the Configuration Page will open.
 - b. Enter your SSO, Country and **if in US select the edit button to change the default email address** (by default, this app will send the email to the Europe IB Team Resource.EuropeanCentralAdminGIB@ge.com). The email to be used in US is usa.ib.verification@ge.com.
3. Start your first Scan.
 - a. Go to the Menu, make the appropriate selection for the job you are performing. **New Install, Upgrade System, or Update System.**
 - b. Enter the applicable Data, Install End Date, System ID & RFS#.
 - c. Click **Scan Bar Codes** and start scanning each barcode on the PL Cards.
 - d. Click the “Email” Button.
 - e. Email is sent to the Default email address, which was set up earlier.

1.8 Required Equipment and Tools



NOTICE

Follow all required safety and PPE procedures customary for your organization when working on this product.

Table 1-1: General Safety Requirements

Item	Description
1	Wear composite safety shoes for foot protection.
2	Wear gloves for hand protection.
3	Wear safety glasses for eye protection.



FERROUS MATERIAL HAZARD!

IF MAGNET IS ENERGIZED, TOOLS, COMPONENTS, AND PARTS REQUIRED FOR THIS INSTALLATION THAT CONTAIN FERROUS MATERIAL WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES.

KEEP ALL FERROUS TOOLS AS FAR FROM THE MAGNET AS POSSIBLE.

Tools to upgrade the system are listed in following table:

Table 1-2: Installation Equipment

Item	GE Part Number	Description
1	5112581	Non-Magnetic Service Tool Kit
2	46-271138G1	Restricted Access Control Kit. Contains two plastic warning signs for posting at site during installation and service activity.
3		Service Methods DVD
4	46-198094P1	Wrist grounding strap
5	---	Plastic or aluminum flashlight
6		Assorted Cable Ties

1.9 Discarded Material Return Policy

The equipment or material being removed by the upgrade have a potential value for service of the installed base. The disposition of such must be controlled by GE. Disposition of the removed material is per GE Medical Systems Policy and Procedures.

1.9.1 Material Policy Overview

1.9.1.1 Returned Material

Per GE Healthcare Policy 3.9: all requested material is to be returned, as directed, intact and promptly to Approved GE Recycling Center. Do not remove or swap any parts from the material to be returned.

1.9.1.2 Disposed Material

Regulations governing the disposal of this equipment or material vary from location to location and from country to country. In order to insure compliance with all local, state, or national environment, health, and safety regulations, dispose of all items per local GEHC policy and procedures and other applicable published guidance. Contact GEHC Recycling Center for assistance.

1.9.2 Harvest Steps



NOTICE

The equipment must be tested prior to the system shutdown/upgrade to allow appropriate Harvest capabilities.

Before going on site:

- Print the proper regional address labeling from DOC2100599 Harvest Return Addresses.
- Verify that you can download the Harvest test procedure from MyWorkshop using DOC1899393.
- If you have not been contacted to test the system or cannot download the document, reach out to the local Harvest team. For all other requests contact:

@HEALTH Global Harvest Team GlobalHarvestTeam@ge.com

Once testing is completed, submit the test record for review. Once these test records are reviewed, they will be processed for electronic signatures.

Package the equipment and follow your regions method of returning harvest equipment.

1.9.3 Harvested Materials

Table 1-3: Harvested Materials For Systems with 5269997 GOC ASSY

System Type for Upgrade	MR450, MR450W, MR750 and MR750W- Test Procedure DOC1899393	
Part Number	Description	Qty
5183547-30	MR DVMR Host	1
5262750	GOC PDU ASSY (Change up to Part Number 5447829)	1
2395043-2	GOC AUDIO BOX	1
2395043	GOC AUDIO BOX	1
5139480-2	8-Port Switch Kit with 4 cushions	1

NOTE: Reuse the packaging that the new GOC or Host PC was shipped in to return the removed GOC or Host PC.

Table 1-4: Harvested Materials For Systems with 5269997-8 GOC ASSY for DV with Z400

System Type for Upgrade	MR450, MR450W, MR750 and MR750W- Test Procedure DOC1899393	
Part Number	Description	Qty
5370415	MR - DV z400 Host	1

NOTE: Reuse the packaging that the new GOC or Host PC was shipped in to return the removed GOC or Host PC.

Table 1-5: Harvested Materials For Systems with 5269997-11 GOC ASSY for DV Z400 with RoHS

System Type for Upgrade	MR450, MR450W, MR750 and MR750W- Test Procedure DOC1899393	
Part Number	Description	Qty
5370415-10	MR - DV MCF z400 Host	1

NOTE: Reuse the packaging that the new GOC or Host PC was shipped in to return the removed GOC or Host PC.

Table 1-6: Harvested Materials For Systems with 5269997-12 GOC ASSY for DV Z420

System Type for Upgrade	MR450, MR450W, MR750 and MR750W- Test Procedure DOC1899393	
Part Number	Description	Qty
5922000	HP z420 Host PC	1

NOTE: Reuse the packaging that the new GOC or Host PC was shipped in to return the removed GOC or Host PC.

1.9.4 Return Kit

If needed, packaging and/or transportation details can be obtained through the local Installation Specialist.

Reuse appropriate containers that new hardware and modules were shipped to site for return of above material.

1.9.5 Deinstallation Procedures

Refer to instructions in this manual for deinstallation procedures of site.

1.9.6 Product Locator Information

Process product locator information for all serialized components removed from the system according to policy. Refer to [Section 1.7](#).

1.10 Upgrade Components

GOC with Dell T5810 Host PC

Chapter 2 GOC Upgrade Procedures

1 GOC Upgrade Procedures

This upgrade procedures provides instructions for upgrading an existing GOC with a 8400, z400, or z420 PC to a GOC with a Dell T5810 PC.

NOTE: For GOC with an 8400 Host PC, the entire GOC must be replaced. For GOC with a z400 and z420 Host PC, either the entire GOC can be replaced or just the Host PC can be upgraded to T5810.

1.1 Power Off Procedure

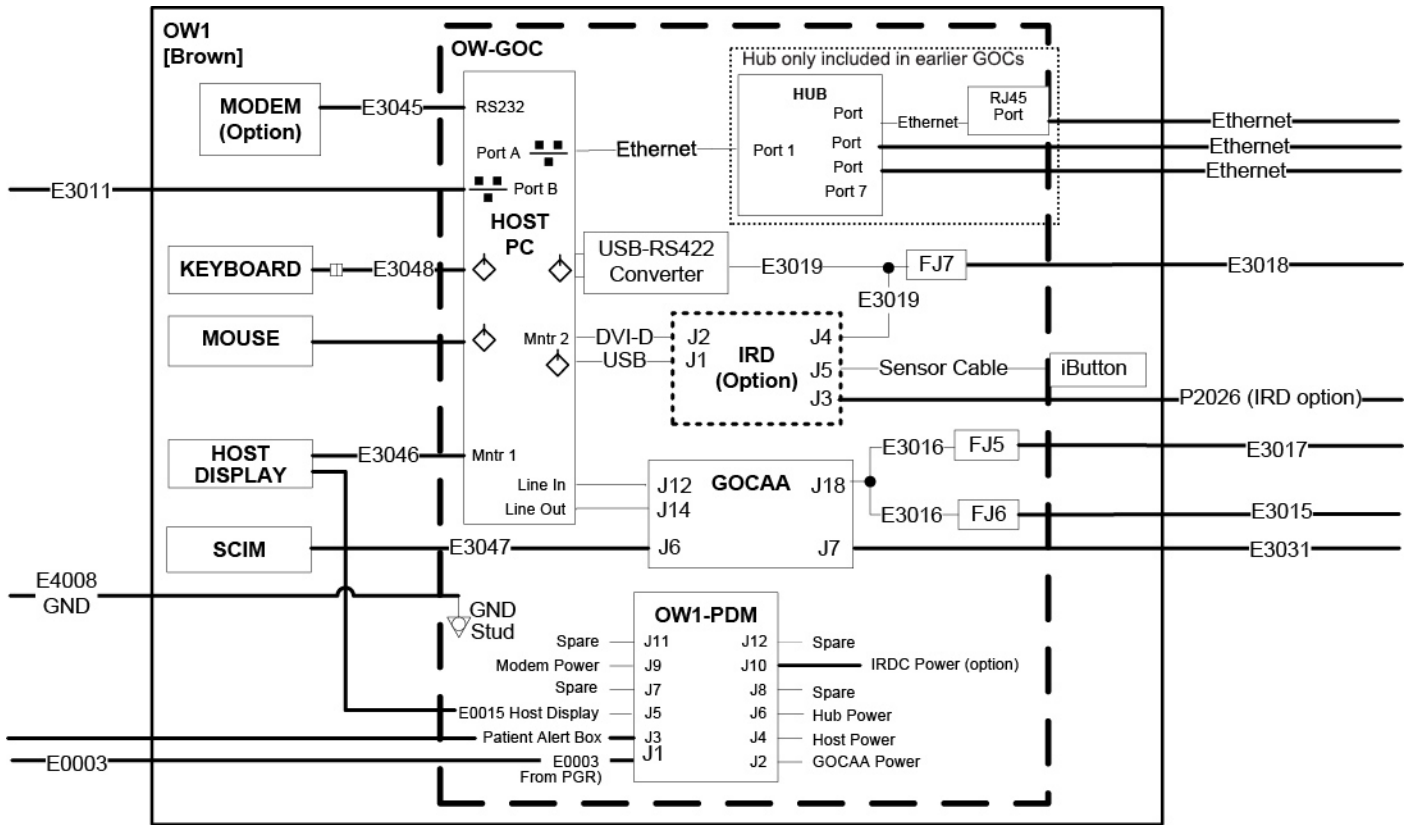
1. Make sure the host PC is powered down.
2. Make sure the LOTO has been performed for the GOC at the PGR cabinet. See the section on LOTO Global Operator Console / Operator Workspace in the *MR Service Safety Manual*, PN 5452735.

1.2 Interconnect Maps

The following sections provide information on cable reuse, component transfer from old GOC to new GOC, and GOC replacement.

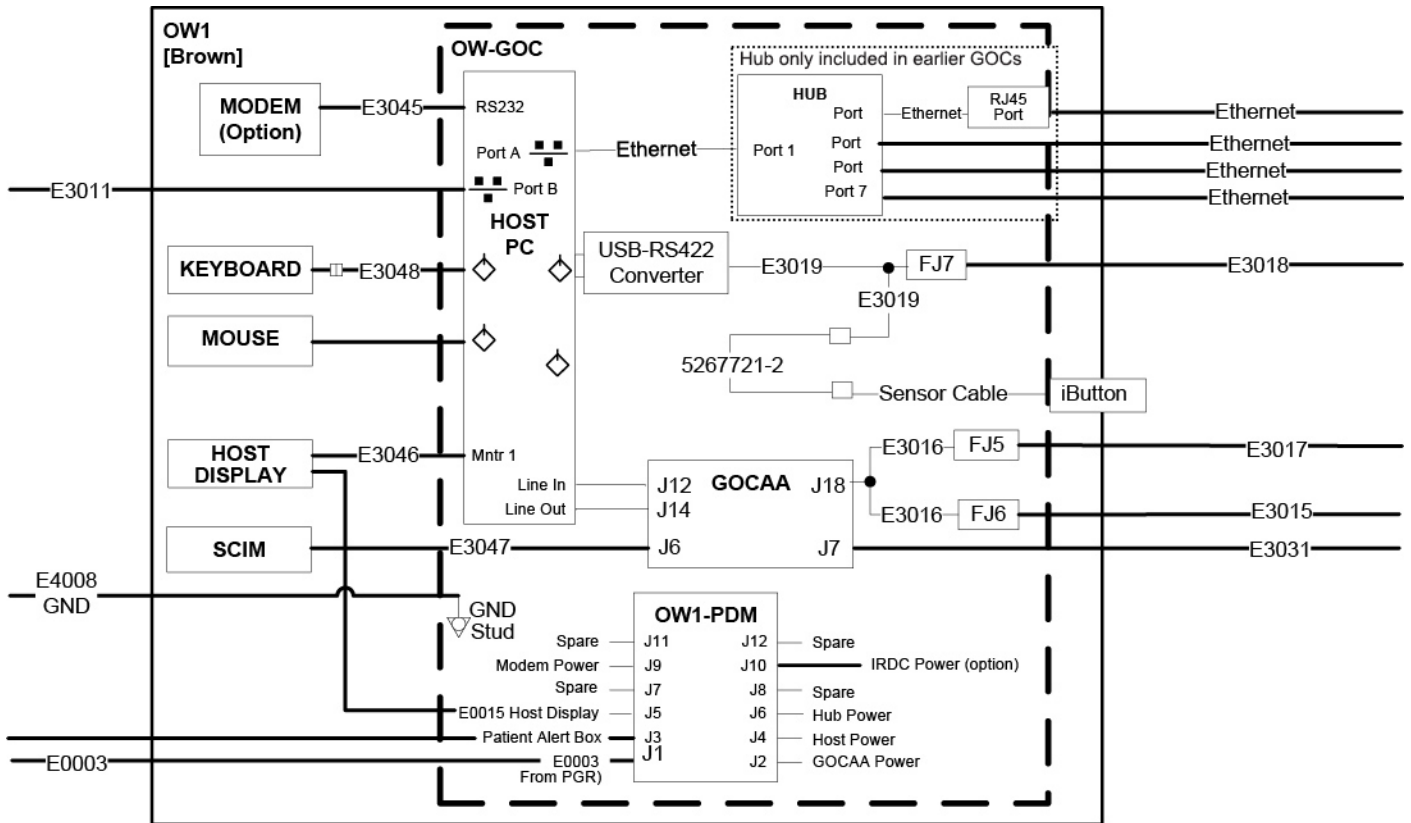
1.2.1 Cable and Component Upgrade for System with IRD

Illustration 2-1: System with IRD



1.2.2 Cable and Component Upgrade for System without IRD

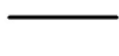
Illustration 2-2: System without IRD



Legend



= GOC to be removed and returned for recycling. Replaced with new GOC.

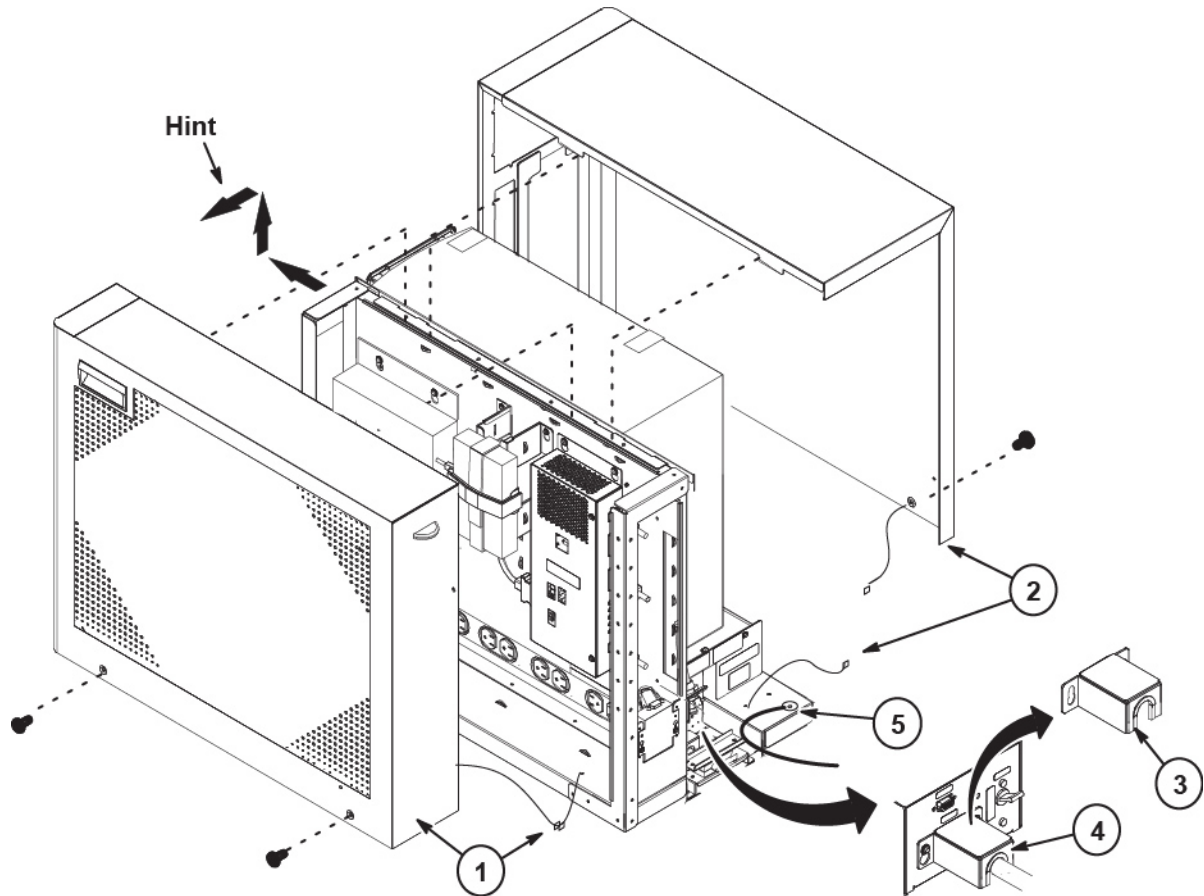


= Cable to be disconnected and reused for new GOC

1.3 Removal Procedure for Existing GOC

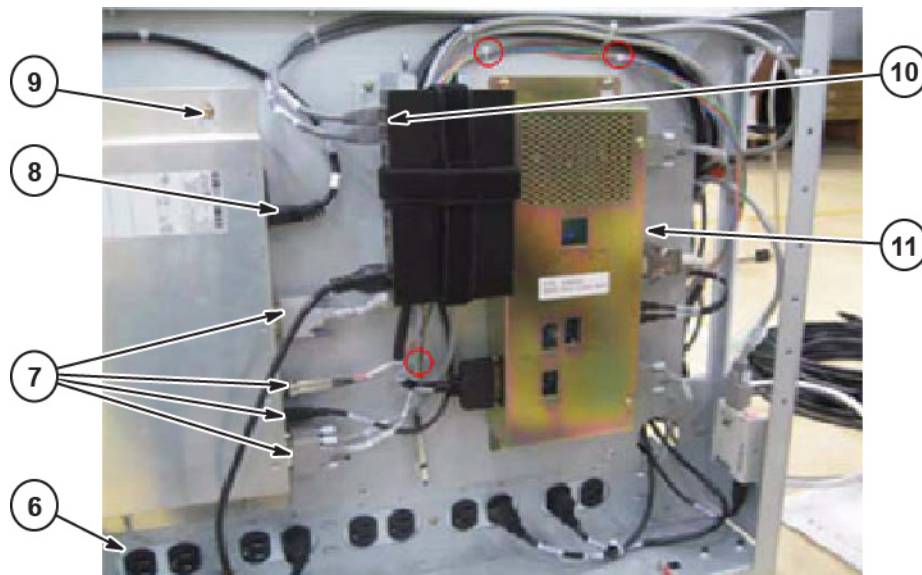
Cables routed to GOC will be reused for connecting to new GOC. Remove cable ties as necessary to make it easier to remove the cables that are routed into the GOC

Illustration 2-3: Remove GOC Covers and Power Cord



1. For right side cover removal:
 - a. Remove two screws.
 - b. To access cover ground cable, carefully move cover forward and then lift up and to the right.
 - c. Disconnect ground cable and remove the cover.
2. For left side cover: Remove the two screws and lift up. Disconnect the ground cable and remove cover.
3. Loosen screws and remove power cable cover.
4. Loosen power cable clamp. Disconnect power cable from J1.
5. Disconnect ground cable.

Illustration 2-4: GOC Module Cable Removal

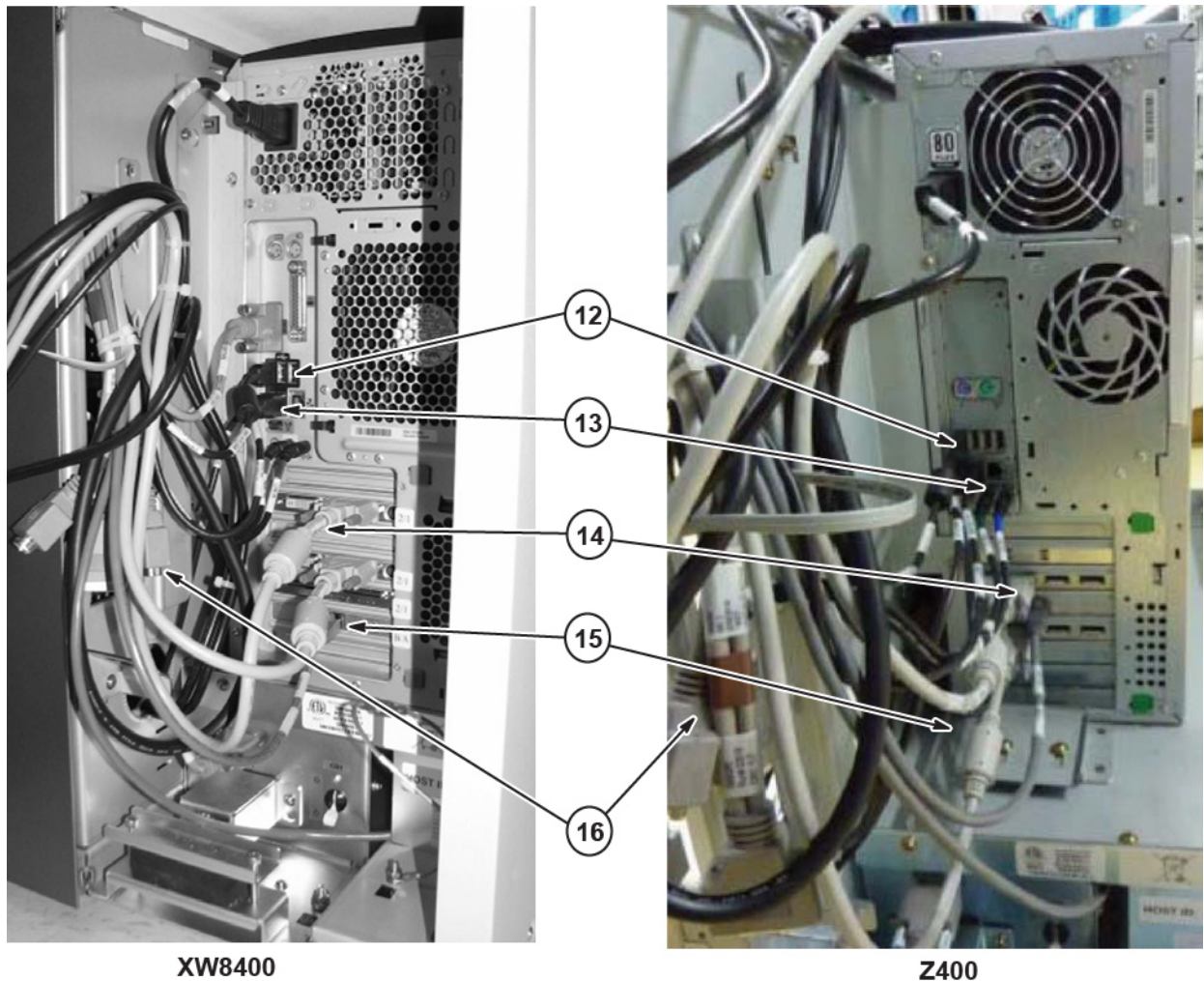


6. Disconnect power cables from GOC PDU.

NOTE: IF IRD Fiber Optic Converter Box option is installed, complete the next three steps.

7. Disconnect all data cables from IRD module.
8. Disconnect power cable from IRD module.
9. Remove the four M5 screws attaching the IRD module to GOC and remove module. The module will be transferred to the new GOC.
10. Label and disconnect all ethernet cables routed to the LAN Hub.
11. Disconnect Runs E3047 and E3031 from J6 and J7 from the GOC AA module.

Illustration 2-5: PC Cable Removal



NOTE: The illustration above would be typical for removal of cables from a z420 host PC.

- 12. Disconnect all cables from the USB ports.
- 13. Disconnect Run E3011 from the ethernet port.
- 14. Disconnect monitor cable Run E3046 from the MNTR1 port.
- 15. If the IRD option is installed, remove the DVI-D cable from the MNTR2 port.
- 16. Disconnect Runs E3015, E3017, and E3018 from FJ6, FJ5, and FJ7.

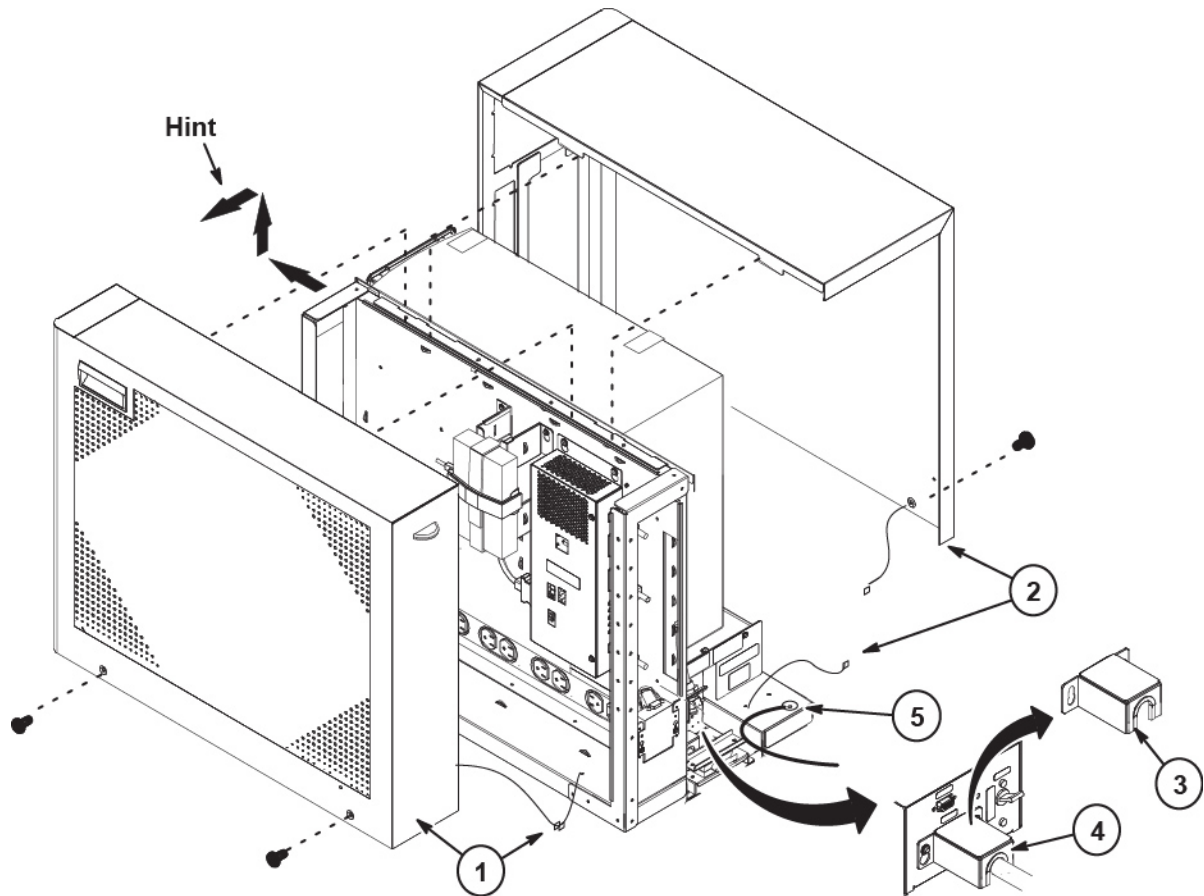
Attach covers to GOC and prepare to return GOC for recycling using the packaging material that the new GOC was delivered to site with.

1.4 Install the New GOC

Carefully unpackage the new GOC and move into place. The packaging materials will be used for return of the old GOC.

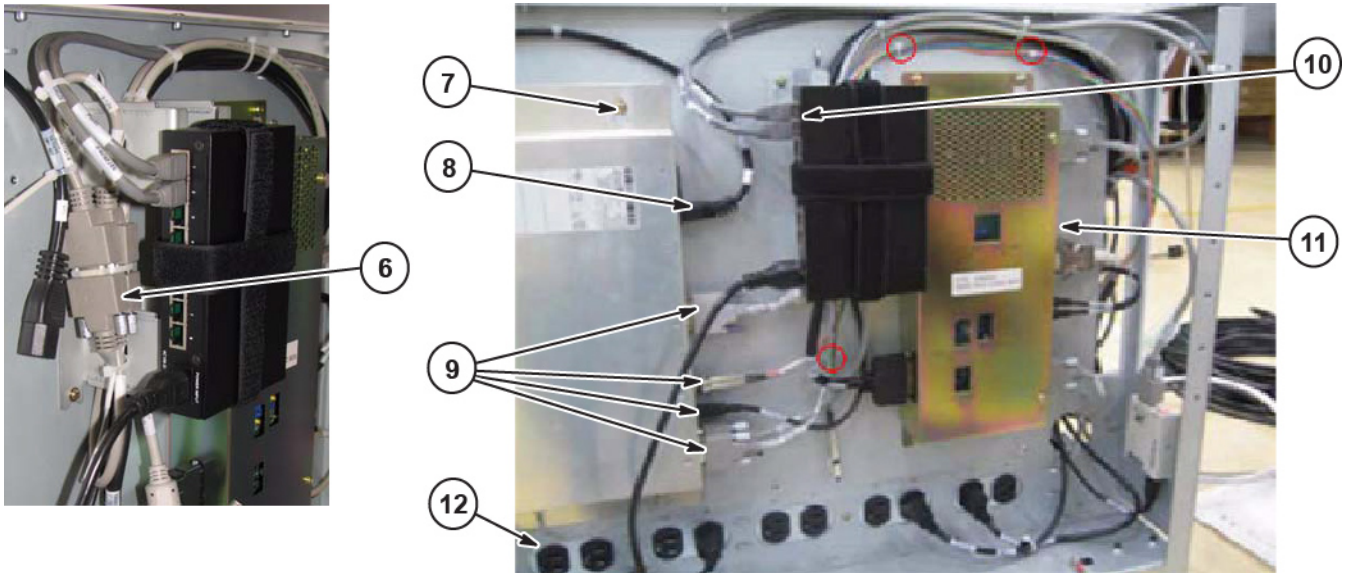
1.4.1 Connect GOC Module Cables

Illustration 2-6: GOC Cover Removal and Power Cable Installation



1. For right side cover removal:
 - a. Remove two screws.
 - b. To access cover ground cable, carefully move cover forward and then lift up and to the right.
 - c. Disconnect ground cable and remove the cover.
2. For left side cover: Remove the two screws and lift up. Disconnect the ground cable and remove cover.
3. Loosen screws and remove power cable cover.
4. Loosen power cable clamp. Route and connect existing power cable E0003 to J1.
5. Connect Run 4008 to GND. make sure the nut securing ground cable is firmly tightened to GND stud. Tighten cable clamp to secure cables in place.

Illustration 2-7: Cable connections to GOC Modules

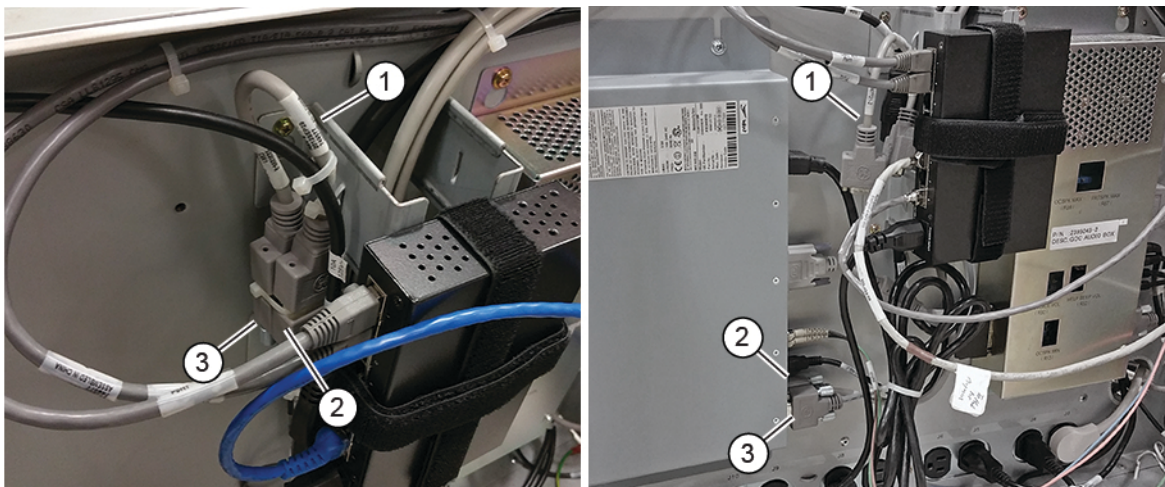


NOTE: Complete the next four steps if:

- IRD Fiber Optic Converter Box option is being transferred from removed GOC
- OR
- Site is being upgraded to IRD from a LED Display.

6. Disconnect both ends of jumper cable 5267721-2 and connect free end of cable Run E3019 to IRD J4 and free end of sensor cable to IRD J5. Jumper cable will be unused.

Illustration 2-8: Jumper Cable Before (left) and After (right)



1	Jumper cable
2	Cable Run E3019 (to IRD J4)
3	Sensor cable (to IRD J5)

7. Install the IRD Fiber Optic Converter Box using the four M5 screws.

8. Connect the power cable to the IRD module.
9. Connect the following cables to the IRD Fiber Optic Converter Box.
 - a. Connect the DVI-D cable from the Host PC to J2.
 - b. Route fiber optic cable Run P2026 to GOC and connect to J3.



WARNING

LASER RADIATION HAZARD!

INVISIBLE LASER RADIATION CAN BE EMITTED FROM DISCONNECTED FIBER-OPTIC CABLES OR CONNECTORS. DO NOT STARE INTO THE BEAM, POINT THE CABLE OR CONNECTOR ENDS AT YOUR EYES, OR VIEW THROUGH OPTICAL INSTRUMENTS.



NOTICE

Electrostatic discharge (ESD) can cause permanent damage to electronic components.

Observe the following ESD precautions:

- Wear a static wrist strap to ensure that any accumulated electrostatic charge is discharged from your body to the ground.
- Create a common ground for the equipment being worked on by connecting the static-free mat, static wrist strap, and peripheral units to that piece of equipment.



NOTICE

The SFP optical transceiver is very sensitive to ESD. SFP latent failures can occur, in dry climates or low relative humidity, surpassing their ESD protection.

Contaminant failures can occur in SFP optics and exposed glass Rx and Tx cable ends. Do not touch the exposed ends on fiber connectors. Leave the dust caps in place until you are ready to install the SFP module. Use the Fiber Optics Cleaning Kit to clean both the SFP and cable end every time they are replaced or swapped from one port to another, or when using a loopback.

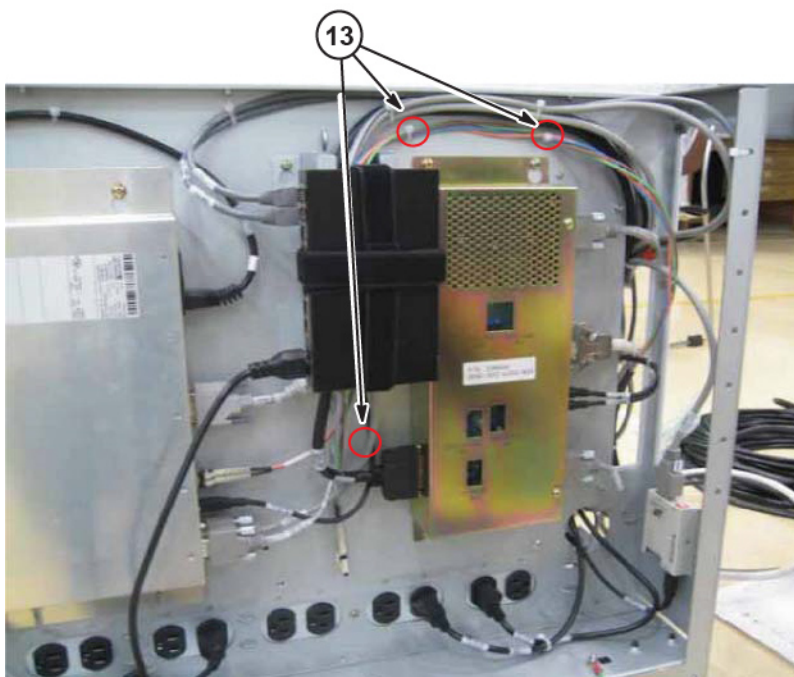
When working with the SFP optical transceiver or SFP Loopback, also observe the following precautions:

- Always use ESD protection when handling the SFP optical transceiver to prevent early failure.
- Always keep a protective cap on unplugged fiber connections.
- The SFP must be shipped in antistatic packaging. Discard any SFP shipped or stored without proper ESD protection.

For more information on handling and cleaning the SFP transceiver refer to Fiber Optic Cable Devices Installation and Replacement.

- c. Connect the USB cable from the Host PC to J1.
 - d. Connect Run E3019 to J4 and the Sensor cable to J5.
10. For GOCs which include an 8-port hub, reconnect all Ethernet cables that were removed to the hub. For newer GOCs without a hub, Ethernet connections can be made directly to the Host PC. If multiple Ethernet connections are required, use a customer-supplied Ethernet switch.
11. Reconnect Run E3047 to J6 and Run E3031 to J7 on the GOC AA
12. Reconnect power cables to PDU Module. Refer to [Section 1.4.2](#)

Illustration 2-9: Securing Fiber Optic Cable



GOC side view

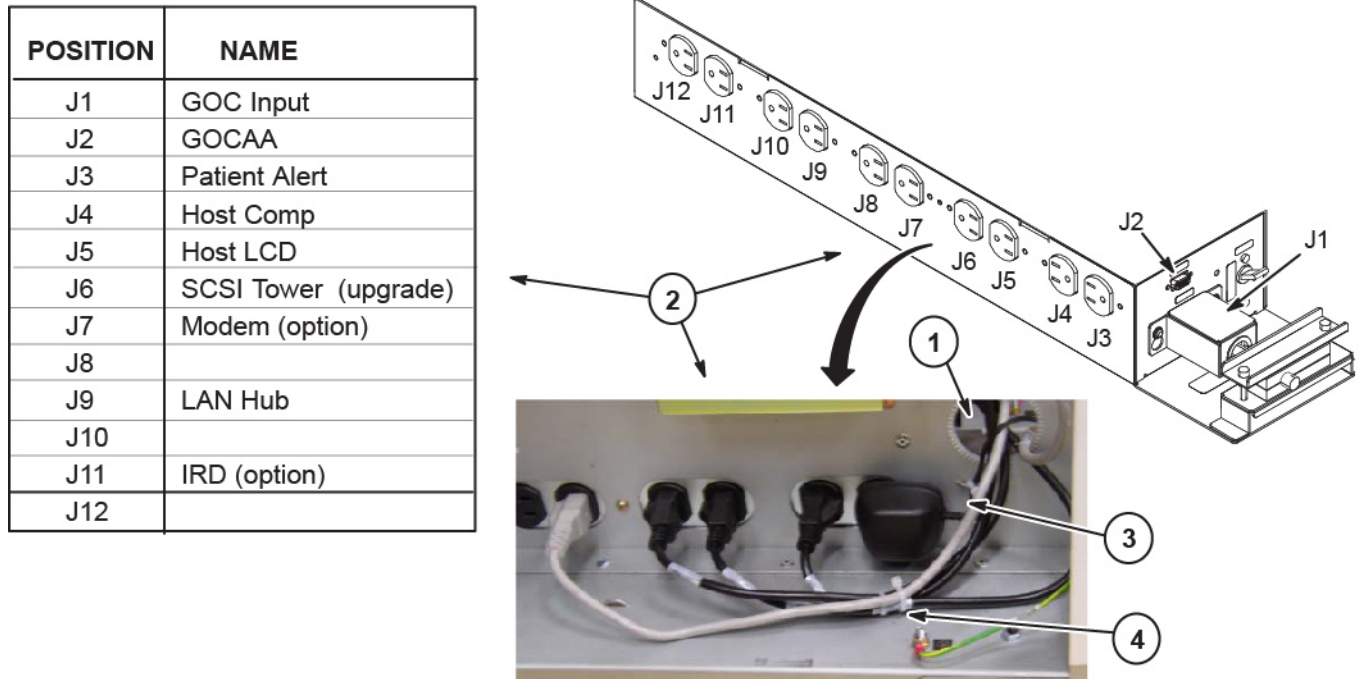


GOC rear view

13. Secure the fiber optic cable with tie wraps at points indicated. make sure the tie wraps are loose as to not damage the optical cable.

1.4.2 Connect GOC PDU Cables

Illustration 2-10: Power Cable Connections

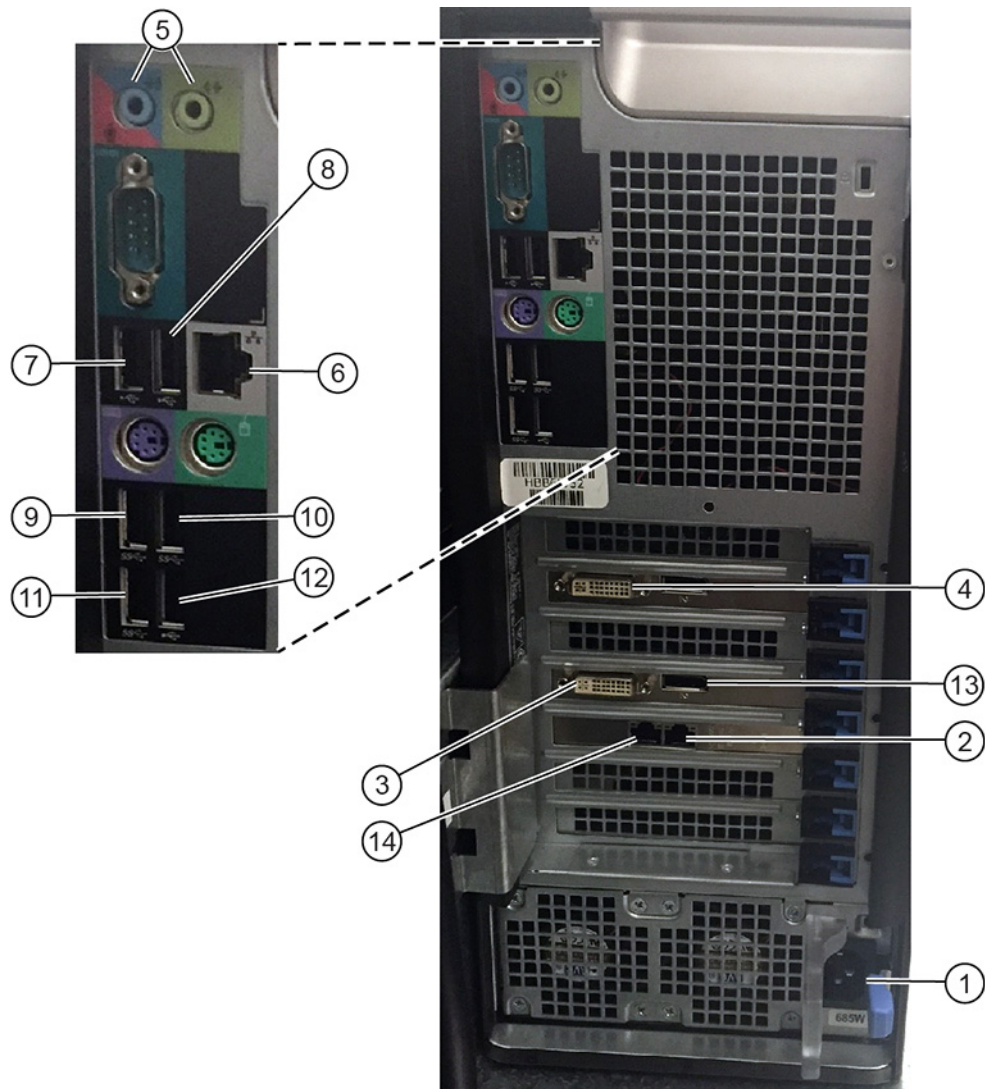


1. Route power cables connected to J3 through J7 through access hole.
2. Connect power cables to PDU module as indicated in table.
3. Route and connect Patient Alert power cable to J3 and tie-wrap at this point.
4. Tie-wrap power cables J4 to J7 at this point.

1.4.3 Connect GOC PC Cables

The connection points of cables routed to the PC are described as follows. Several of the cables shown are delivered with the GOC and are preconnected.

Illustration 2-11: PC Area Connections



NOTE: USB 3.0 ports are marked with an SS symbol (items 9, 10, and 11 in [Illustration 2-11](#)). Other USB ports (items as 7, 8, and 12) are USB 2.0.

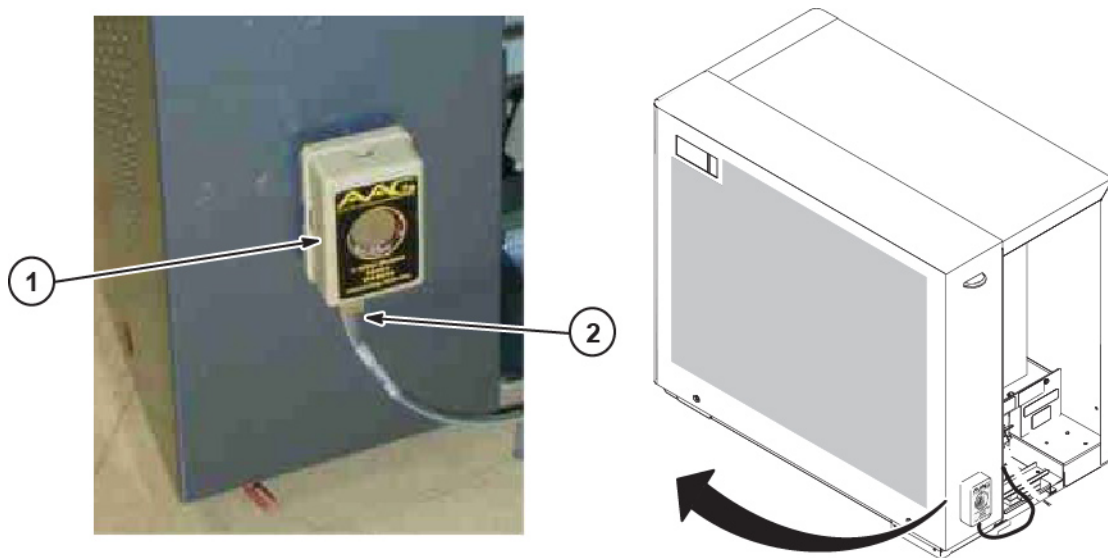
1. See [Illustration 2-11](#). Route and connect PC power cord.
2. Run E3011 MUST be connected to the RIGHT side Ethernet port.
3. If the IRD option is installed, connect DVI-D cable to MNTR2 port PCI5.
4. Monitor cable Run E3046 will be connected to MNTR1 port PCI2.
5. Route and connect the Line In and Line Out cables from the GOCAA.
6. For GOCs which include an 8-port Ethernet hub, route and connect Ethernet cable from GOC hub.

For newer GOCs without a hub, this port is available for direct connection to a service laptop, site network, or AW option. If multiple Ethernet connections are required, use a customer-supplied Ethernet switch.

7. Route and connect USB cable from IRD to USB 2.0 port.
8. Route and connect USB-RS422 Converter cable to USB 2.0 port.
9. **(For Systems with ICE)** Route and connect Run E3037 USB cable to USB 3.0 port.
10. Not used in this system. Connect optional BrainWave accessory cable.
11. Connect mouse (Run E3048) to USB 3.0 port.
12. Connect keyboard extension cable.
13. Not used in this system. Connect optional BrainWave video cable.
14. Item 14 in [Illustration 2-11](#) has no connection.
15. Connect Runs E3015, E3017, and E3018 to FJ6, FJ5, and FJ7 as marked.
16. Dress and secure cables.

1.4.4 Restore GOC Covers

Illustration 2-12: Install GOC Covers



1. Attach iButton Sensor assembly to back of GOC Cabinet at this location using Dual Locking Velcro Tape (2341779 & 2341779-2).
2. Make sure cable (5267721) is connected to Sensor.

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Chapter 3 Power On Procedures

1 Power On Procedures

1.1 Remove LOTO

To remove LOTO, see the section on LOTO Global Operator Console / Operator Workspace in the *MR Service Safety Manual*, PN 5452735.

1.2 Finalization Procedures

NOTE: The T5810 PC is only compatible with DV25.1 and later software.

1. Power on Host PC.
2. Complete Software Load procedures referring appropriate Service Methods Documentation.
3. Update eLicense site with new Host ID and Secure Service Access (SSA) site with new Service License ID.
4. Perform a Check Scan and hand over system to customer.
5. As **root user**, enter the following series of commands to run the USB mouse fix.

su

<password>

usbmousefix

6. Reboot the Host PC when prompted.

Illustration 3-1: USB Mouse Fix

```
{sdc@t11}[102] su
password:
{sdc@t11 sdc}# usbmousefix
#####
usbmousefix: Start: Thu Feb 23 23:39:16 CST 2017
#####
USB Output:
Bus 001 Device 001: ID 1d6b:0002 Linux Foundation 2.0 root hub
Bus 002 Device 001: ID 1d6b:0002 Linux Foundation 2.0 root hub
Bus 003 Device 001: ID 1d6b:0001 Linux Foundation 1.1 root hub
Bus 004 Device 001: ID 1d6b:0001 Linux Foundation 1.1 root hub
Bus 005 Device 001: ID 1d6b:0001 Linux Foundation 1.1 root hub
Bus 006 Device 001: ID 1d6b:0001 Linux Foundation 1.1 root hub
Bus 007 Device 001: ID 1d6b:0001 Linux Foundation 1.1 root hub
Bus 008 Device 001: ID 1d6b:0001 Linux Foundation 1.1 root hub
Bus 001 Device 002: ID 0930:6545 Toshiba Corp. Kingston DataTraveler 102/2.0 / HEMA Flash Drive 2 GB / PNY Attache 4GB Stick
Bus 006 Device 002: ID 048d:c014 Logitech, Inc. Corded Workstation Mouse
Bus 006 Device 003: ID 03f0:0024 Hewlett-Packard KU-0316 Keyboard
#####
USB Device Details are captured in /usr/local/bin/install/log/usbdevices.log
Parsing usb devices for mouse:
#####
list of applicable USB mouse quirks:
0x046d:0xc014:0x0400
#####
Checking grub conf for existing quirks ...
kernel /boot/vmlinuz-2.6.32-573.3.1.el6.x86_64 ro root=UUID=e879c84a-2399-4d52-b7a5-4c440db2b50a nomodeset rd_NO_LUKS KEYBOARDTYPE
epc KEYTABLE=us LANG=en_US.UTF-8 rd_NO_MD SYSFONT=latarcyrheb-sun16 crashkernel=auto rd_NO_LVM biosdevname=0 rd_NO_DM rhgb quiet fips=1 aud
it=1 usbhid.quirks=0x046d:0xc014:0x0400
#####
Removed old quirks from grub.
#####
Saving /etc/grub.conf.*quirk before applying new quirks...
-rw-r--r-- 1 root root 970 Jan 18 10:35 /etc/grub.conf.b4quirk
-rw-r--r-- 1 root root 977 Feb 23 23:39 /etc/grub.conf.newquirk
#####
Successfully added quirks for usb mouse IDs below:
0x046d:0xc014:0x0400
#####
kernel /boot/vmlinuz-2.6.32-573.3.1.el6.x86_64 ro root=UUID=e879c84a-2399-4d52-b7a5-4c440db2b50a nomodeset rd_NO_LUKS KEYBOARDTYPE
epc KEYTABLE=us LANG=en_US.UTF-8 rd_NO_MD SYSFONT=latarcyrheb-sun16 crashkernel=auto rd_NO_LVM biosdevname=0 rd_NO_DM rhgb quiet fips=1 aud
it=1 usbhid.quirks=0x046d:0xc014:0x0400
Done: Thu Feb 23 23:39:18 CST 2017
Reboot required for changes to take effect.
#####
usbmousefix: End: Thu Feb 23 23:39:18 CST 2017
#####
```

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